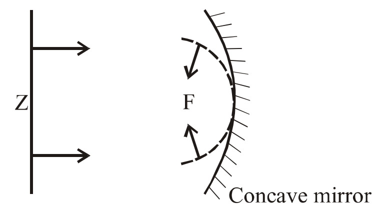
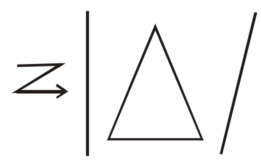
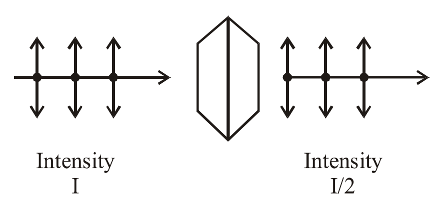
Hints & Solutions

1. The wave theory was not readily accepted because light could travel through vacuum and it was felt that a wave would always require a medium to propagate from one point to the other. These two facts contradict each other.
2. Light wave consists of electric and magnetic field which are time and space varying field.
3. A wavefront is locus of all the particles oscillating in same phase.
5. Two source are coherent then the phase difference ϕ at any point will not change with time and we will have a stable interference pattern.
6. In the phenomenon of interference, energy in conserved but it is redistributed. In interference of light, the enegy is transferred from the region of destructive interference to the region of constructive interference. The average energy being always equal to the sum of the energies of the interfering waves. Thus the phenomenon of interference is in complete agreement with the law of conservation of energy.
9. By using white light, the central maxima will be white while the fringes closest on either side of central fringe is red and farthest will appear blue.
11. Thus ΔQ will be small if the diameter of the objective is large. This implies that the telescope will have better resolving power if 'a' is large. It is for this reason that for better resolution, a telescope must have a large diameter objective.
12. From equation $d_{min} = \dfrac{1.22\lambda}{2\mu\sin\theta}$

    It is to be noted that it is not possible to make sin θ larger than unity. Thus, we see the resolving power of a microscope is basically determined by the wavelength of light used.
14. The phenomenon of polarisation is based on the fact that light waves are transverse E.M. wave.
30. When e^- move from ground state to excited state. Then it remain 10^{-8} sec in excited state and come to ground state.
31. 

32. Two separate sources are two independent source meanes two non coherent source, so we obtain average intensity (uniform intensity).
33. Time between object means incident wave front & image (emergent wave front) will be same for every point hight will take more time to cross prism but upper side it will take less time so cover more distance as compare to base of prism towards base of prism.



34. In interference intensity same for all bright band but in diffraction. It decrease when we move away from central maxima.
35. Due to one slit we obtain diffraction pattern on screen.
36. Resolving power = $\dfrac{2\mu\sin\beta}{1.22\,\lambda}$
38. 

39. According to malus law

    $I' = \dfrac{I}{2}\cos^2\theta$

    when θ = 90 I' = 0

    when θ = 0 $I' = \dfrac{I}{2}$

    when 0 < θ < 90°

    0 < I' < I/2

40. Light has a wavelength of about half a micrometer. If it encounters an obstacle of about this size, it can bend around it and can be seen on the other side.
General life object are generally much bigger than wavelength of light.

41. The size reduces by half according to relation $\text{size} \simeq \frac{\lambda}{d}$ $\therefore$ the intenstity increases four fold.

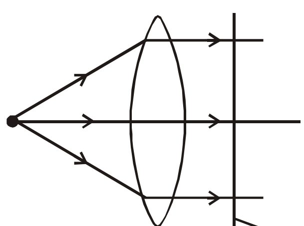
42. Waves diffracted from the edge of the circular obstacle interfere constructively at the centre of the shadow producing a bright spot.

43. The speed of light in vacuum is a universal constant independent of all the factors listed and anything else. In particular note the surprising fact that it is independent of the relative motion between source & observer.

44. Wave speed does not depend on nature of source it depends on property of medium, also it is independent of direction of propagation for isotropic medium.

45. Interference of the direct signal received by the antenna with the signal reflected by the passing air craft.

46.



47. Planar as a small area on the surface of a large sphere is nearly planar.

48. According to Huygen's principle, each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all direction with the speed of the wave. These wavelets emanating from the wavefront. are usually referred to as secondary wavelets & if we draw a common tangent to all these spheres, we obtain the new position of the wavefront at a later time.

49. Doppler shift can be expressed as

$$\frac{\Delta f}{F} = \frac{-V_{radial}}{C}$$

Where V_{radial} is the component of the source velocity along the line joining the observer to the source relative to the observer.

50. For a given frequency, intensity of light in the photon picture is determined by the number of photons crossing an unit area per unit time.

51. $Z_F = a^2/\lambda$. The spreading due to diffraction is smaller compared to the size of the beam. When distance between slit & screen is much smaller than Z_F.
$Z_F = a^2/\lambda$

52. Width of pinhole = 10^3Å = 1000 Å
wavelength of sun light ranges from 4000Å to 8000Å
wavelength λ < width of the slit Hence, light is diffracted from the hole. Due to diffraction from the slight the image formed on the screen will be different from the geometrical region.

53. Intensity $\propto \frac{1}{\text{area}}$
hole size$\uparrow$ width of central maxima$\downarrow$ Intensity$\uparrow$